

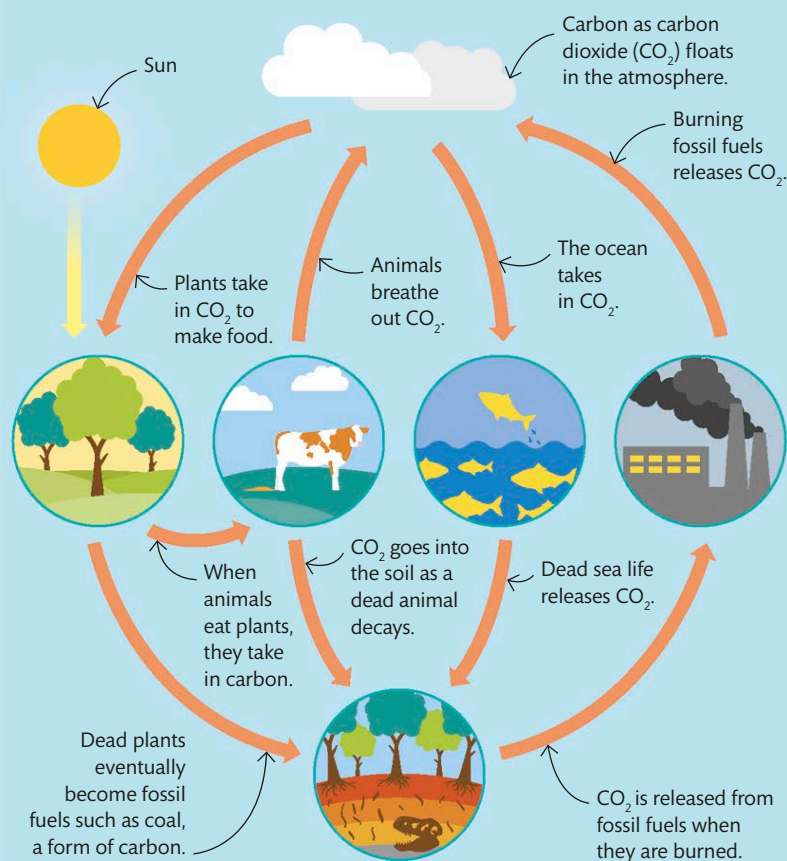
UNLOCKING THE PAST

DATING FOSSILS

All matter is made up of atoms (see page 177). For each element, there are several different forms, called isotopes, each with a different number of neutrons. Some isotopes are radioactive, which means they break apart, or decay. When animals or plants die, they leave behind many isotopes in their bodies, including carbon isotopes. By measuring the levels of these isotopes in a fossil, scientists can discover when the organism died.

THE CARBON CYCLE

The element carbon, most often in the form of CO_2 , constantly passes between living things and the air, soil, and water whenever living things breathe, eat, or die. This is the carbon cycle. When something dies, it stops taking in carbon, so the mix of carbon isotopes is fixed. As time passes, some of the isotopes decay, so measuring their concentration allows scientists to date when the animal or plant died.



Dinosaur fossils

These fossils found in Sahatsakhan, Thailand, belong to a type of plant-eating dinosaur called a sauropod. Back in the lab, scientists will study the uranium isotopes in the fossils to calculate how many millions of years ago the dinosaur died. Dinosaur remains are too old for carbon dating, as the carbon isotopes will have fully decayed.